AR/VR simulations for Stoma Siting

Abstract

A survey paper reviewing and summarising recent research in the field of AR/VR simulations for stoma siting.

1 Introduction

In collaboration with Singhealth, our project aims to enhance patient understanding and mental preparedness for stoma siting using Augmented Reality/Virtual Reality (AR/VR) simulations to address anxiety, promote informed decision-making, and optimise overall patient experience.

This survey reviews and summarises recent research in the field of AR/VR simulations and the use of AR/VR simulations for patient preparation.

2 Classification

From the papers selected for review, common areas that the research aims to investigate or resolve have been identified. We have used the following areas as classification:

1. Mixed Reality (MR) for patient education
2. MR for medical practitioners
3. MR's ability to perform simulations

3 Mixed Reality (MR) for patient education

3.1 Virtual Reality–Based Intervention to Reduce Preoperative Anxiety in Adults Undergoing Elective Surgery A Randomized Clinical Trial (Lim Zhe Rui)

Pre-surgical stress can affect surgical outcomes and increase the chances of post-operative complications. This paper[5] explores how VR interventions can be used to reduce preoperative anxiety and improve post-operation outcomes for adults in a generalised surgery setting.

The experiment conducted was a prospective, single-blind randomised clinical trial on adult patients who were scheduled for their first elective surgery procedure under general anaesthesia within the next 2 to 4 weeks, spoke Cantonese, and were grade I or II of the American Society of Anesthesiologists Physical Status Classification System. Half of the participants were selected for the control group and the rest for the intervention group. The intervention group was shown a VR-based intervention that simulated the perioperative process. Data was collected through surveys conducted at various stages.

The experiment showed that preoperative anxiety and other outcomes like preparedness and post-operative anxiety were improved in the intervention group. However, there was no real change in pain levels or postoperative stay length.

One limitation identified was COVID-19, which greatly reduced the sample size due to surgeries being cancelled, and may have influenced the results in unknown manners. Due to the nature of the intervention, its allocation also could not be hidden from the patients in the intervention group, which may have biased them towards giving positive answers. Another limitation was the experiment being limited to a single clinic, which may limit the generalizability of the findings as standards of care may differ between clinics.

3.2 Augmented Reality for Perioperative Anxiety in Patients Undergoing Surgery (Roy Julien Michael)

Preoperative anxiety affects 60-80% of surgical patients, and can extend hospital stay, slowing recovery. Preoperative education reduces anxiety, however, traditional methods are limited. This paper [11] hypothesizes augmented reality (AR) technology can provide an effective solution, offering patients an immersive walk-through of their surgical experience.

The study had a randomized clinical trial including 95 patients assigned randomly to standard preoperative instructions or additional AR education. The AR application allowed patient interaction with the environment.

Significant improvements found in anxiety levels for the AR group. The AR group experienced decreased anxiety from screening to post-intervention with no change in the control group. Persisting until surgery, the AR group maintaining lower anxiety levels, as the control group's anxiety increased. 71.4% of the AR group reported enjoyment of the experience. Reduced anxiety was present regardless of enjoyment of the AR experience.

Limitations include: 1. Limited generalization from focusing on a singular surgeon and institution, 2. Patients survey response decreased in post-operative surveys, 3. AR intervention focuses on the day-of experience and not the post-operative experience, 4. Lacked the necessary sample size.

3.3 Virtual reality for preoperative patient education: Impact on satisfaction, usability, and burnout from the perspective of new nurses (Roy Julien Michael)

Traditional education methods create significant stress for new nurses, require extensive training, and result in inconsistent quality across different nurses. Inconsistency affects both patient care and contributes to nurse burnout. This paper [14] hypothesizes virtual reality (VR) technology can improve both education process' and reduce nurse burnout.

The study conducted a comparative experiment involving 20 new nurses and 80 patients for plastic surgery. Each nurse provided education to four patients, two using traditional methods and two using VR technology. The VR system covered the entire hospitalization process. Evaluated using three key metrics: satisfaction (using a 7-item questionnaire), usability (using System Usability Scale (SUS) and After-Scenario Questionnaire), and burnout (using the Maslach Burnout Inventory).

The study had a relatively small sample size, (20 nurses) and was limited to a single department. Additionally, patients with visual impairments or claustrophobia were excluded, introducing bias. The evaluation relied heavily on subjective measures, questionnaires rather than objective indicators. Finally, practical implementation challenges were identified, including the high costs of VR equipment and content development for an advancing field.

Ultimately, results demonstrated significant improvements across multiple measures. The VR education group showed statistically higher satisfaction scores compared to traditional methods. Usability also favored the VR system, though the overall SUS score of 68 was only average. Regarding burnout, emotional exhaustion decreased and personal accomplishment improved. However, depersonalization showed no significant change.

3.4 Intraoperative use of virtual reality decreases anxiety during surgery under local anaesthesia in the head and neck region (Roy Julien Michael)

The challenge of high patient anxiety during surgeries performed under local anesthesia is addressed in this paper[17]. High anxiety affects patient comfort and surgeons attentions, which can compromise surgeries. Research has demonstrated benefits of audio-visual distraction methods. The authors sought to understand VR's effectiveness as a distraction.

Conducting a non-randomized prospective study with 95 patients, the VR group saw calming animated virtual environments while being operated on. The study included multiple surgical sites to determine if certain areas were more suitable for VR intervention.

The results demonstrated significant benefits of VR for patient anxiety. Lower intraoperative anxiety scores compared to the control group and high patient satisfaction. 86.6% recommending VR use during surgery and 71.7% reporting they would be less anxious about future surgeries if VR was available. The study found no differences in pain levels between groups. The surgical site location also did not affect anxiety levels, suggesting VR's versatility across different procedures.

The study lacked randomization and the variation in surgical procedures between groups introduced bias. Different factors potentially affected patient experience. The study also revealed practical challenges: patients refused VR use, three patients required early removal of headsets, and some patients reported fitting issues or preference for real landscapes. Additionally, previous VR experience was not quantified.

3.5 Training Application for Ostomy Patients in Virtual Reality (Zhang Mengjie)

This paper[9] examines the creation of a virtual reality (VR) training program for ostomy patients. The tool replicates postoperative care settings, enabling patients to rehearse stoma management in a virtual setting. The study indicates favorable results, encompassing enhanced patient confidence and less anxiety. Nonetheless, the application predominantly focuses on postoperative care rather than preoperative education. This gap underscores the necessity for VR technologies that explicitly address the stoma siting procedure.

This research presents an interactive StomaVR designed to assist patients in ostomy care. The VR application concentrated on three primary procedures: 1) evacuating the ostomy bag, 2) replacing the ostomy bag, and 3) substituting the plate to which the bag is affixed. The processes are taught in several steps to facilitate instruction. The application can be used from a relative's perspective, allowing the patient to interact with the avatar and use body tracking.

The study's efficacy was examined by a questionnaire evaluating presence, engagement, realism, interface quality, interactions, and audio. The findings indicated that the application was effective; however, the study's sample size of nine participants, aged 24 to 59 years, with merely two lacking prior experience, implies that the results may not reliably reflect the efficacy of VR applications across a varied patient demographic over a prolonged duration.

3.6 Virtual and augmented reality in critical care medicine: the patient's, clinician's, and researcher's perspective (Niik Seah)

Patients are often stressed, anxious, pained, and delirious in ICU environments. Traditional methods of patient education lack interactivity and engagement. This paper[4] seeks to evaluate patients' perspectives on the applications of VR/AR in critical care medicine.

The study evaluated the implementation of VR systems in various uses: relaxation therapy for anxiety and stress reduction, distraction for pain management, gaming to encourage physical activity, and education and learning support for clinicians.

The study concluded a significant reduction in stress, anxiety, and discomfort in patients, reporting high patient satisfaction and acceptance of VR in stress relief and aftercare. Clinicians reported improved adherence to guidelines and confidence in procedures as well.

However, the studies conducted were small in scale and lacked control groups, reducing reliability. There was also a lack of standardization in hardware, software, and study designs utilised, making it difficult to compare results across trials.

3.7 Evaluating Mixed Reality as a Tool for Patient Education of Facial Fractures (Niik Seah)

Two-dimensional computed tomography (2DCT) is the primary modality for patient education on facial fractures. However, complex anatomical information proves challenging for 2DCT to present to patients understandably, affecting their ability to make informed decisions about their treatment. This paper[6] seeks to evaluate Mixed Reality (MR) as a tool for patient education on facial fractures, where three-dimensional mixed reality (3DMR) goggles present a possibility as a more immersive and intuitive method through the visualizations of fractures in a 3D interactive format.

A prospective, randomised, crossover study comparing 2DCT and 3DMR as formats for education was conducted. Two groups of 25 were recruited; one viewed the 2DCT format first, followed by 3DMR, while the other viewed 3DMR first, then 2DCT. After each viewing, participants filled up a survey on self-assessment, information recall, and preference to test their understanding, knowledge retention, and preference respectively.

Participants preferred the 3DMR format over 2DCT, reporting enhanced understanding with no significant differences in information recall.

Limitations of the study include the small sample size, limiting the experiment's generalizability. In addition, questions in the information recall survey were not validated, which may have potentially influenced the experiment's accuracy.

3.8 Bridging the gap: evaluation of preoperative patients' education by comparing expectations and real-perioperative surgical experiences: a mixed-methods descriptive cross-sectional study (Niik Seah)

Preoperative patient education is essential in reducing anxiety and closing the gap between expectations and experiences. However, the mismatch between information provided by the healthcare provider and the specific concerns and expectations of patients results in gaps in the understanding and satisfaction of patients. This paper[1] seeks to evaluate preoperative patient education by comparing patients' expectations and real-perioperative surgical experiences.

A mixed-methods descriptive cross-sectional study was conducted with 65 patients, with preoperative and postoperative interviews conducted to assess patient expectations versus real experiences.

It was concluded that effective communication and interactive education using practical demonstrations reduce anxiety by aligning surgical expectations and outcomes.

Two limitations were identified, including the small scale of the experiment which limits the generalizability of the results and the self-reported data which introduces potential biases.

3.9 The effect of using virtual reality technology on anxiety and vital signs before surgery in patients undergoing open heart surgery (Cui Shenyi)

The paper [2] explores the effectiveness of using VR to lower patients' preoperative anxiety before open heart surgery. Preoperative anxiety has been known to affect surgical outcomes and patient recovery. For example, anxiety has been linked to delayed wound healing and an increased risk of post-operative complications. Moreover, anxiety can lead to an increase in heart rate and blood pressure, which can be harmful for patients with heart conditions.

The study split the 60 participants undergoing open-heart surgery into two groups: one that watched VR videos and one that watched ordinary videos. The videos were 4 minutes long, explaining the operation and surgical process; the participants were then examined using the Spielberger State-Trait Anxiety Inventory (STAI) before and after the video intervention.

The study found that the VR group's mean anxiety score dropped from 55.8 to 38.6, while the ordinary video group's reduced from 58.33 to 45.13. There were also improvements in other vital signs, such as heart rate and blood pressure, with greater effects in the VR group. Despite the promising results, the study was conducted on a minimal sample size and would benefit from being tested on a larger scale.

3.10 Virtual Reality, Augmented Reality, and In Vivo Exposure Therapy: A Preliminary Comparison of Treatment Efficacy in Small Animal Phobia (Cui Shenyi)

This paper[16] compares the efficacy of Vivo Exposure Therapy (iVET) to the effectiveness of Visual Reality Exposure Therapy (VRET) and Augmented Reality Exposure Therapy (ARET). iVet refers to a form of exposure therapy where the patient is placed directly in front of their feared object. However, in some cases, it raises some ethical concerns and is often met with patient reluctance. Ninety-one participants were exposed to their phobia of spiders and cockroaches, and their behavioural avoidance score was recorded.

The study found that all three treatment modalities (iVET, VRET, ARET) were similarly effective in reducing fear, anxiety, and avoidance. All three are viable, effective treatments that prepare patients and help them with their phobias. However, some limitations of the study were that it was primarily conducted with women, which may limit the generalizability of the results.

4 MR for medical practitioners

4.1 Virtual Reality for Immersive Education in Orthopedic Surgery Digital Twins (Zhang Mengjie)

The integration of virtual reality (VR) with surgical digital twins (SDTs) could significantly enhance medical education and procedural preparation. This paper[10] presents SurgTwinVR, a VR application that mimics a surgical operating room down to the last detail, allowing users to practice their skills in a safe and realistic setting. An all-inclusive surgical scene, including the surgeon, anatomical components, surgical equipment, and the operating area, may be simulated in this first-of-its-kind virtual dynamic 3D environment.

The virtual operating room's 3D model was created using laser scans and high-resolution photographs, generating a 3D point cloud by amalgamating RGB-D pictures captured by ceiling-mounted cameras. The surgical drill was monitored using a minuscule infrared-reflective marker and an optical tracking system. The surgeon and instrument were captured at a rate of 30 frames per second, and surgeons were presented with geometrically flawless point clouds with a holographic appearance.

However, the study has several limitations such as realism tradeoffs, and high cost and technical complexity. The system requires some specific equipment or cameras to capture the actual procedures and the point cloud representation may not fully replicate the tactile and sensory aspects of the real-world surgery.

4.2 Acceptance of Virtual Reality Simulation Training for Stoma Care by Healthcare Providers: A Pilot Questionnaire Study After Viewing Prototype Imagings (Zhang Mengjie)

This paper[13] developed two virtual reality (VR) simulation prototypes from the viewpoints of ostomates and caretakers utilizing modeled stomas. The project team initiated the "OSTrain-VR Project," a stoma self-care training program for ostomates. This method allows experienced ostomates to communicate their advanced stoma self-care procedures with novices using VR imaging, facilitating simulations at their convenience and frequency. Furthermore, seasoned nurses can impart immersive experiences to less experienced healthcare personnel or the relatives of stomates.

The researchers created VR simulation movies for ostomate self-management and caregiver-supervised stoma care. About three-minute VR movies were shot and produced to show stoma pouch

removal, skin cleansing, and pouch application from different angles. This study conducted a questionnaire survey among healthcare professionals working in various facilities to solicit their feedback. More than 90 per cent of participants supported its clinical application in all cases. Interest was equally split between "definitely want to use" and "want to use with free access."

This study may have a small sample size, selection bias, and no control group. Researcher omitted 2D-video software as a comparison group. The subsequent phase could involve conducting a prospective clinical study that employs VR-based training for patients undergoing stoma formation surgery.

4.3 Augmented reality in the operating room: a clinical feasibility study (Cui Shenyi)

The paper [7] highlights the potential for surgeons to use Augmented Reality (AR) to assist them in orthopaedic surgery. It identifies the problem that existing navigation systems (technological tools used to guide surgeons during procedures) are too costly. It aims to evaluate the possibility and practicality of using cheaper AR technology, such as the HoloLens, as a compact alternative for navigation.

The AR headset prototype provided surgeons with 3D patient-specific anatomical models during surgery. The AR technology was designed to allow surgeons to manipulate the models with hand gestures and voice commands. It was tested by 13 orthopaedic surgeons over 25 different procedures and rated highly for image accuracy and quality. However, the ratings for the accuracy of voice commands and gesture control suffered.

Other limitations included an ergonomic concern about the devices and environmental factors that may impair voice commands. These small-scale studies were done for very short procedures, so more trials are needed to determine their effectiveness. Despite the limitations, the study shows promise.

4.4 Augmented Reality Visualization for Laparoscopic Surgery (Yuan Hao)

This paper [8] addresses key challenges in laparoscopic surgery by developing an augmented reality (AR) system to enhance surgical visualization. Traditional laparoscopy relies on 2D imaging, forcing surgeons to infer depth through motion or physical probes, while existing stereo laparoscopes lack natural head-motion parallax and viewpoint alignment. Additional issues include a narrow field of view (requiring frequent camera adjustments) and hand-eye coordination mismatches between instrument movements on-screen and in reality.

The proposed solution integrates a 3D AR system combining video-see-through head-mounted displays (HMDs), structured light-based laparoscopes (to capture depth and color data), and tracking systems to align virtual imagery with patient anatomy. This setup aims to merge real-time 3D surgical visuals with the surgeon's natural perspective, improving depth perception and spatial awareness.

Experiments evaluated two prototypes: one using manually digitized pre-operative 3D models for AR-guided tasks, and another employing real-time structured light to generate dynamic 3D meshes. Results showed surgeons successfully completed tasks (e.g., target piercing in foam models) with reduced cognitive load, and the custom 3D laparoscope and lightweight HMD improved depth perception and usability. However, delays in depth extraction, calibration errors, and hardware rigidity highlighted technical limitations. Tests also relied on simplified anatomical models, lacking clinical validation.

Key limitations include slow depth processing, calibration inaccuracies, and hardware bulk. Future work proposes faster depth capture technologies, miniaturized flexible laparoscopes, adaptive depth acquisition for complex anatomies, and integration with pre-operative imaging (e.g., MRI/CT) for holistic visualization. The study underscores AR's potential to transform laparoscopic surgery while outlining critical technical hurdles for further research.

4.5 Augmented reality for the surgeon: Systematic review (Yuan Hao)

This systematic review [18] explores the application of augmented reality (AR) in surgical settings to address challenges in orthopedic surgery, such as surgeons' difficulty translating 2D fluoroscopy images into 3D anatomy—a process that increases cognitive strain and procedural errors—as well as risks from frequent radiation exposure. Existing robotic systems, while precise, are criticized for disrupting workflows and requiring invasive setups. Persistent inaccuracies in implant placement (e.g., pedicle screws, acetabular cups) and alignment during osteotomies compound these issues, alongside training gaps for novices in complex tasks like tumor resections or needle placements.

To tackle these problems, experiments tested augmented reality (AR) solutions. AR-guided pedicle screw placement using systems like HoloLens and hybrid optical-electromagnetic tracking reduced reliance on fluoroscopy, while AR-generated panoramic X-rays improved alignment accuracy in cadaver osteotomies. Tablet-based AR projected 3D tumor margins in pig models, enhancing resection precision, and AR simulations boosted novice success rates in phantom-based training tasks. Trauma applications, such as AR-guided femoral guide-wire placement, further demonstrated reduced procedural errors.

Evaluation showed AR-guided pedicle screws achieved 97% accuracy in models, radiation exposure dropped by 50–70% in vertebroplasty, and AR-enabled tumor resections achieved 90% safe margins (vs. 71% conventionally). Surgeons reported lower cognitive strain with AR navigation, and novices using AR training saw success rates rise to 80% (vs. 35% without). Despite these advances, technical hurdles like registration errors from patient movement, parallax distortions, and bulky AR hardware (e.g., HoloLens) remain. Small preclinical sample sizes, hardware flaws (e.g., short battery life), and workflow disruptions from manual calibration or marker dependency also limit clinical adoption. While AR holds significant promise, addressing these barriers is critical for its integration into routine practice.

In conclusion, while AR holds promise for enhancing surgical precision and workflow efficiency, significant barriers remain. Hardware limitations, ergonomic design flaws, and a lack of clinical validation in diverse settings hinder widespread adoption. Future advancements will require miniaturized devices, seamless integration with preoperative imaging, and rigorous testing in live, complex surgical environments to fully realize AR's potential in modern medicine.

4.6 Applicability of augmented reality in orthopedic surgery – A systematic review (Yuan Hao)

This paper[12] examines challenges in orthopedic surgery that augmented reality (AR) could mitigate, including surgeons' difficulty translating 2D fluoroscopy into 3D anatomy—a process that heightens cognitive strain and error risks—alongside harmful radiation exposure from frequent imaging. While robotic systems improve precision, they disrupt workflows and lack adaptability. Persistent inaccuracies in implant placement (e.g., pedicle screws, acetabular cups) and alignment during osteotomies compound these issues, as do training gaps for novices in complex tasks like tumor resections or needle placements.

To address these, researchers tested AR solutions: systems like HoloLens and hybrid optical-electromagnetic tracking improved pedicle screw placement accuracy while reducing fluoroscopy reliance. AR-generated panoramic X-rays enhanced alignment in cadaver osteotomies, and tablet-based AR projected 3D tumor margins in pig models, achieving safer resections. Trauma applications, such as AR-guided femoral guide-wire placement, reduced procedural errors, while AR simulations boosted novice success rates in phantom-based training tasks.

Results demonstrated AR's potential: 97% accuracy in pedicle screw placement, 50–70% reduced radiation exposure, and 90% safe tumor margins (vs. 71% conventionally). AR-trained novices achieved 80% success rates in simulations (vs. 35% for controls), and surgeons reported reduced cognitive strain during navigation.

However, limitations include technical hurdles like registration errors from patient movement, parallax distortions, and bulky AR hardware. Methodological constraints—small preclinical sample sizes, over-reliance on phantoms—and hardware flaws (e.g., HoloLens' vergence conflicts, short battery life) hinder clinical adoption. Workflow disruptions from manual calibration and marker dependency further challenge integration. While AR shows transformative promise in precision, radiation reduction, and training, resolving these barriers is essential for its widespread use in practice.

5 MR's ability to perform simulations

5.1 On the Use of Virtual Reality for Medical Imaging Visualization (Lim Zhe Rui)

Medical imaging visualisation requires high image quality and data control, and would benefit from 3D visualisation and taking advantage of medical information formats such as DICOM. Under these circumstances, this paper[15] assesses the current use of VR in clinical contexts and aims to identify the current limitations and possible future extensions.

The paper identifies features that would be useful in the context of clinical practice, such as 3D drawing and labelling, and evaluates existing applications in the market. Commonly used development tools and frameworks, such as Unity and MeVisLab, were also used to develop prototypes using publicly available datasets to evaluate how difficult they were to use.

The paper concludes that most existing solutions do not have sufficient performance to be used in a real-life surgical setting and are currently best suited for education and training. In terms of development tools and frameworks, game engines were identified as the best choice for development as they are the most accessible, are heavily supported, and versatile in the features offered. The main limitations identified for existing solutions are real-time performance, high precision, and minimal latency.

5.2 Augmented Immersive Telepresence Experience with a Virtual Body (Lim Zhe Rui)

This paper[3] explores how a virtual body could be used to improve immersion in a telepresence experience where one or more parties are participating virtually, with other parties present physically.

The experiment tested whether users preferred having a virtual body, and whether having a virtual body in a telepresence situation increased place illusion (feeling like you were in another location) and co-presence. Participants sat through a telepresence scenario which was conducted twice for each participant, once with virtual arms and one without. Robotic arms were chosen to avoid the issue of the uncanny valley and to avoid having to create specific arm models for each participant. An additional action was performed in each session, such as throwing a plush rat at the participant, to elicit movement from the participant and gauge an implicit sense of presence. Data was gathered through post-session and post-experiment questionnaires. An introductory study was conducted before a confirmatory study with a greater number of participants.

The data showed that the introductory study and confirmatory study had seemingly contradictory results, with the introductory study showing an increase in Place Illusion, co-presence and preference. The confirmatory study did not show an increase in co-presence, and there was only weak evidence of preference. After further investigation and analysis, the paper concludes that the presence of a virtual body improving immersion depends on the preferences of the individual. The naturalness of the virtual body, affected by factors such as the size and position of the virtual arms compared to the participant's actual body, and its interactions with the physical environment could impact immersion.

The paper identifies a few limitations. The stimuli chosen for eliciting motion from the participant could break their presence in the simulation and could also cause an accommodation effect. The main metric used for measuring hand motions, integration, is also susceptible to noise.

6 Conclusion

In recent years, there has been increased interest and research into the use of MR in simulations to educate patients on surgical procedures and improve surgical outcomes. The results so far have been promising, but many of the studies are limited in either sample size or location, and the majority of them have been in general settings. There has been limited studies focused on the specific procedure of stoma siting, especially for reducing preoperative anxiety, which gives us the opportunity to expand on this neglected area through the development of a prototype to evaluate MR's effectiveness in this field.

Furthermore, many studies have also reported limitations in MR simulations, either due to software or hardware limitations inhibiting the field as a whole due to decreased immersion. In the future, there will likely be more research conducted into discovering how to overcome these limitations with novel solutions.

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